

## ✓ Step-by-Step Instructions

### ◆ Step 1: Verify Installations

1. Open CMD as administrator.
2. Run:

```
bash

java -version
hadoop version
```

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Confirm Java version `1.8.x` and Hadoop version like `3.2.4`.

### ◆ Step 2: Start Hadoop Services

```
bash

start-all.cmd
```

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Ensure **DataNode**, **NodeManager**, **ResourceManager**, **NameNode** windows appear and stay running.

Check running services:

```
bash

jps
```

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### ◆ Step 3: Access Web UIs

- HDFS: `http://localhost:9870`
- ResourceManager: `http://localhost:8088`

### ◆ Step 4: Create HDFS Input Directory

```
bash

hadoop fs -mkdir /input
```

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### ◆ Step 5: Prepare Input Text File

1. Create a `.txt` file, e.g., `page1.txt`, with some words (including duplicates).
2. Copy the file to HDFS:

```
bash

hadoop fs -put "C:\your\local\path\page1.txt" /input
```

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## ◆ Step 6: Setup Eclipse Java Project

1. Create a new Java Project: `MapReduceWordCount`
2. Create a new package: `com.mapreduce.wordcount`
3. Create a class: `WordCount.java`
4. Add external JARs from:

- `HADOOP_HOME/share/hadoop/` folders:
  - `common`
  - `common/lib`
  - `hdfs`
  - `hdfs/lib`
  - `mapreduce`
  - `yarn`
  - `client`

## ◆ Step 7: Write and Paste Java Code

Use this sample `WordCount.java` class (or copy from video description):

```
java Copy Edit

package com.mapreduce.wordcount;

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {

    public static class TokenizerMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
```

#### ◆ Step 8: Export as JAR

- Right-click project → Export → Java → JAR File
- Save as: `WordCountMapReduce.jar`

#### ◆ Step 9: Run the JAR in Hadoop

```
bash Copy Edit
hadoop jar "C:\path\to\WordCountMapReduce.jar" com.mapreduce.wordcount.WordCount /input/page1.txt
```

#### ◆ Step 10: View Output

```
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hadoop fs -cat /output/part-r-00000
```

Expected Output:

```
nginx Copy Edit

Ganesh      2
laptop      1
Maharashtra 2
subscribers 1
technical   1
windows     1
```

## ❖ Final Notes

- Make sure your input directory, file name, and class/package names match exactly.
- Use `hadoop fs -ls /output` to verify output folder contents.
- Delete output folder before re-running:

```
bash Copy Edit
hadoop fs -rm -r /output
```

Would you like me to provide the Java code as a downloadable file or help troubleshoot a specific step?